ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Chemical Reactions and Equations** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** one or more new substances	Chemical changes form new substances.
2	**(b)** combination reaction	Two reactants form one product.
3	**(d)** blue	Hydrated copper(II) ions are blue.
4	**(d)** reactants	Reactants are written on the left.
5	**(a)** decomposition reaction	One reactant gives two products.
6	**(c)** white	Barium sulphate is a white insoluble salt.
7	**(b)** CaO	Quick lime is calcium oxide.
8	**(d)** displacement reaction	Fe displaces Cu.
9	**(d)** yellow	Golden-yellow solid.
10	**(b)** $\mathrm{Ca(OH)_2}$	Slaked lime is calcium hydroxide.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Mass can neither be created nor destroyed in a chemical reaction (1); Total mass of reactants equals total mass of products (1).	1+1
12	A single compound splits into two or more simpler substances (1); $\mathrm{CaCO_3 \longrightarrow CaO + CO_2}$ (1).	1+1
13	Slow destruction of a metal by air, moisture or acids (1); Rust ($\mathrm{Fe_2O_3 \cdot x\,H_2O}$) (1).	1+1
14	Magnesium + oxygen (1); gives magnesium oxide (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Count atoms: 1 Na, 2 Cl on the left, 1 Na, 1 Cl on the right (1); Put 2 before NaCl and 2 before Na (1); $\mathrm{2\,Na + Cl_2 \longrightarrow 2\,NaCl}$ (1).	1+1+1
16	Combination (1); Two reactants form a single product (1); Exothermic (heat is released) (1).	1+1+1
17	Nail becomes covered with a reddish-brown deposit (1); Blue colour of the solution fades (1); Solution turns pale green (1).	1+1+1
	OR: Left: 2 N, 2 H; right: 1 N, 3 H (1); Put 2 before NH3 and 3 before H2 (1); $\mathrm{N_2 + 3\,H_2 \longrightarrow 2\,NH_3}$ (1).	1+1+1
18	Decomposition (1); One compound gives two products (1); Needs heat, so thermal decomposition (1).	1+1+1
	OR: Burns with a dazzling white flame (1); White powder is formed (1); Heat and light are given out (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Burns with a dazzling white flame (1); $\mathrm{2\,Mg + O_2 \longrightarrow 2\,MgO}$ (1); Combination reaction (1); Product is magnesium oxide (1); To remove the oxide layer (1).	1+1+1+1+1
	OR: Slow attack on a metal by moisture and air (1); Needs air (oxygen) and moisture (1); Rust is $\mathrm{Fe_2O_3 \cdot x\,H_2O}$ (1); Prevention: painting, oiling (1); Galvanisation, alloying (1).	1+1+1+1+1
20	Brown fumes are seen (1); $\mathrm{2\,Pb(NO_3)_2 \longrightarrow 2\,PbO + 4\,NO_2 + O_2}$ (1); Thermal decomposition (1); Brown gas is nitrogen dioxide (1); Residue is yellow lead oxide (1).	1+1+1+1+1
	OR: Oxidation of fats and oils in food (1); Occurs in air due to oxygen (1); Food has bad smell and taste (1); Prevention: antioxidants (1); Nitrogen flushing and airtight containers (1).	1+1+1+1+1

Mistakes to avoid:

- Changing the subscripts (formulae) instead of the coefficients while balancing; $\mathrm{H_2O}$ must not be written as $\mathrm{H_2O_2}$ to balance oxygen.
- Writing $\mathrm{Mg + O_2 \longrightarrow MgO}$ without balancing, or $\mathrm{MgO_2}$ for magnesium oxide.
- Calling every reaction in which heat is given out 'decomposition'; identify the type from the reactants and products (one product = combination).
- Forgetting that in a displacement reaction the displacing metal must be **more reactive**; $\mathrm{Cu + FeSO_4}$ gives no reaction.

- Mixing up oxidation and reduction: gain of oxygen is oxidation, loss of oxygen is reduction.
- Writing that rusting needs only oxygen; it needs **both** oxygen and moisture. Slaking of lime is exothermic, not endothermic.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Chemical Reactions and Equations** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** double displacement reaction	Exchange of ions.
2	**(a)** reddish-brown	Copper metal is reddish brown.
3	**(a)** dissolved in water	Aqueous solution.
4	**(b)** exothermic	Exo means outwards.
5	**(d)** fats and oils	Oxidation of fats/oils spoils taste.
6	**(d)** gaseous state	g stands for gas.
7	**(b)** endothermic	Endo means inwards.
8	**(a)** air and moisture	Both oxygen and water are needed.
9	**(d)** souring of milk	Sour milk contains new substances.
10	**(a)** exothermic	Combustion releases heat.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A more reactive element displaces a less reactive element from its compound (1); $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ (1).	1+1
12	Oxidation of fats and oils in food (1); It gives a bad smell and taste (1).	1+1
13	Equation with equal number of atoms of each element on both sides (2).	2
14	An insoluble solid formed in a reaction in solution (1); $BaSO_4$ (white) or PbI_2 (yellow) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Left: 1 Mg, 2 O; right: 1 Mg, 1 O (1); Put 2 before MgO then 2 before Mg (1); $2\,Mg + O_2 \longrightarrow 2\,MgO$ (1).	1+1+1
16	Displacement (1); Zinc displaces hydrogen from the acid (1); Zinc is more reactive than hydrogen (1).	1+1+1
17	Burning of fuels like LPG/coal (1); Respiration (1); Reaction of quick lime with water (1).	1+1+1
	OR: Left 2 O, right 1 O (1); Put 2 before H2O and 2 before H2 (1); $2\,H_2 + O_2 \longrightarrow 2\,H_2O$ (1).	1+1+1
18	Double displacement (1); Ions are exchanged (1); White precipitate of $AgCl$ (1).	1+1+1
	OR: Photosynthesis (1); Thermal decomposition of limestone (1); Electrolysis of water (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Bubbles are seen (1); $Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$ (1); Displacement (1); Gas is hydrogen (1); It burns with a pop sound (1).	1+1+1+1+1
	OR: Exothermic: heat released (1); Endothermic: heat absorbed (1); Examples of exothermic: burning of LPG, respiration (1); Examples of endothermic: photosynthesis (1); Thermal decomposition of limestone (1).	1+1+1+1+1
20	Cathode: hydrogen (1); Anode: oxygen (1); $2\,H_2O \longrightarrow 2\,H_2 + O_2$ (1); Ratio is 2 : 1 (1); Type: electrolytic decomposition (1).	1+1+1+1+1
	OR: Hissing sound and the beaker becomes hot (1); $CaO + H_2O \longrightarrow Ca(OH)_2$ (1); Product is slaked lime (1); Combination reaction (1); Used for whitewashing walls (1).	1+1+1+1+1

Mistakes to avoid:

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- Writing $Mg + O_2 \longrightarrow MgO$ without balancing, or MgO_2 for magnesium oxide.
- Calling every reaction in which heat is given out 'decomposition'; identify the type from the reactants and products (one product = combination).
- Forgetting that in a displacement reaction the displacing metal must be **more reactive**; $Cu + FeSO_4$ gives no reaction.
- Mixing up oxidation and reduction: gain of oxygen is oxidation, loss of oxygen is reduction.
- Writing that rusting needs only oxygen; it needs **both** oxygen and moisture. Slaking of lime is exothermic, not endothermic.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Chemical Reactions and Equations** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** galvanisation	Galvanisation prevents rusting.
2	**(c)** $Fe_2O_3 \cdot x\,H_2O$	Rust is hydrated iron(III) oxide.
3	**(d)** combination reaction	$CaO + H_2O \longrightarrow Ca(OH)_2$
4	**(d)** white	AgCl is a white solid.
5	**(a)** hydrogen	Active metals liberate hydrogen from acids.
6	**(a)** thermal decomposition	Heat is the energy supplied.
7	**(b)** nitrogen	Nitrogen is unreactive.
8	**(b)** 'gives' or 'yields'	The arrow points from reactants to products.
9	**(c)** photolytic decomposition	Light energy is used.
10	**(a)** CuO	Copper(II) oxide is black.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Painting or oiling (1); Galvanisation (zinc coating) (1).	1+1
12	Two or more substances combine to form a single product (1); $C + O_2 \longrightarrow CO_2$ (1).	1+1
13	Oxidation is gain of oxygen (1); Reduction is loss of oxygen (1).	1+1
14	Reaction in which heat is released (1); Burning of coal or respiration (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Right has 2 Cl, left has 1 Cl (1); Put 2 before HCl (1); $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1).	1+1+1
16	Decomposition (1); One reactant breaks into two products (1); Oxygen gas is released (1).	1+1+1
17	Change in state (1); Change in colour (1); Evolution of gas or change in temperature (1).	1+1+1
	OR: Balance H: 2 before H2O (1); Balance O: 2 before O2 (1); $CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$ (1).	1+1+1
18	Combination (1); Two compounds join to give one product (1); White fumes of NH_4Cl are formed (1).	1+1+1
	OR: $2\,H_2O \longrightarrow 2\,H_2 + O_2$ (1); Hydrogen at cathode (1); Oxygen at anode (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Combination: two or more react to give one product (1); $C + O_2 \longrightarrow CO_2$ (1); Decomposition: one compound gives two or more products (1); $2\,H_2O \longrightarrow 2\,H_2 + O_2$ (1); $2\,AgCl \longrightarrow 2\,Ag + Cl_2$ or $2\,H_2 + O_2 \longrightarrow 2\,H_2O$ as second examples (1).	1+1+1+1+1
	OR: White AgCl turns grey (1); $2\,AgCl \longrightarrow 2\,Ag + Cl_2$ (1); Photochemical decomposition (1); Silver metal is formed (1); Used in black and white photography (1).	1+1+1+1+1
20	Combination: $CaO + H_2O \longrightarrow Ca(OH)_2$ (1); Decomposition: $CaCO_3 \longrightarrow CaO + CO_2$ (1); Displacement: $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ (1); Double displacement: $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2\,NaCl$ (1); Redox: $CuO + H_2 \longrightarrow Cu + H_2O$ (1).	1+1+1+1+1
	OR: Oxidation: gain of oxygen (1); Reduction: loss of oxygen (1); H_2 is oxidised (1); CuO is reduced (1); H_2 is the reducing agent (1).	1+1+1+1+1

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